

ME2 — ALFRED: Voice Controlled Smart Device

Always-on keyword + intent recognition at the edge · no cloud round-trip

3 October 2026 · Edward Vincent Duero · Tiny Conformer · Hugging Face · 1× A100 · Raspberry Pi 5

Model

Mic · 16 kHz 40-bin log-Mel	
Tiny Conformer · 4 × 96-dim blocks full-context · attention pooling	
Intent head	Slot CTC head
In-scope gate	Raspberry Pi 5

Item	Value
Parameters / weights	0.936 M · 4.55 MB ONNX

Validation on the Raspberry Pi 5

Item	Value
Keyword / intent acc	90.1 % / 52.5 %
False-accept rate	6.2 % (1 / 16 OOS)
Latency p95 / RTF	12.4 ms / 0.0063
Runtime	onnxruntime · 2 thr · 543 MB peak

Dataset

Item	Value
Source	HF benchmark @ da92a79...ed006 + supplement @ c376ada...d25b5
Hours / utts	23.73 h / 35,810 prepared
Speakers	622 documented
Labels	19 intents · 6 slots · OOS

Training on the A100 cluster

Item	Value
Cluster	1 × NVIDIA A100 · CUDA:0
Objective	CE + 0.5 CTC + 0.5 BCE
Optimiser	AdamW · TPE HPO (6/model)
Steps / loss	35 epochs · batch 64 · seeds 231/947

To Be Submitted

GitHub repository	github.com/vincedevs/AI231-ME · public · project: ME2/ · MIT code
Dataset location	HF airimonda/ai231-me2-voice-commands @ da92a79...ed006 HF vincedevs/alfred-vcm-supplement @ c376ada...d25b5 · public
A100 cluster	1 × NVIDIA A100 · CUDA:0 · 2.44 h · seeds 231 and 947
Model weights	Tiny Conformer ONNX · 4.55 MB · ME2/alfred/assets/models/vcm/ artifact 21c7b228fd63-20559251ca67 · checksummed · terms documented

What the reviewer will look for

#	Item	Status
1	Repo public, one-command reproduction	DONE
2	Dataset licensed and citable (DOI)	DONE
3	Training logs + final checkpoint committed	DONE
4	Pi 5 latency reproduced by the posted script	DONE
5	Held-out test set, unseen speakers	DONE
6	Baseline of comparable size compared	DONE